Course: Computer Vision

Course Code: ITA0506

Slot: D

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Submitted By

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CO4_AT1_DATA INTERPRETATION

1. Real-Time Detection System

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Comparison

Viola-Jones: Fast but has lower accuracy.

YOLO: Good accuracy with moderate speed.

Deep Learning: Highest accuracy but slowest.

YOLO should be selected because it provides 92% accuracy with moderate speed, giving a good balance between detection accuracy and latency.

For strict latency constraints, Viola-Jones is faster, but its 78% accuracy may be insufficient. Deep Learning is more accurate but may cause unacceptable delays.

Conclusion: YOLO is the best practical choice for a real-time surveillance system where both speed and acceptable accuracy are important.

2. Robust Object Detection

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Comparison

Viola-Jones

Normal: 80%

Occlusion: 60%

Low light: 55%

Lowest overall robustness.

YOLO

Normal: 90%

Occlusion: 75%

Low light: 70%

Better robustness than Viola-Jones.

Deep Learning

Normal: 95%

Occlusion: 85%

Low light: 80%

Best performance under all conditions.

Deep Learning should be selected because it achieves the highest accuracy in normal, occluded, and low-light conditions.

Trade-off: It generally requires more computational resources and may have higher latency and cost than the other methods.

Conclusion: Deep Learning provides the best overall robustness, while YOLO may be preferred when speed and resource limitations are important.

3. Optical Flow Analysis

Method A

Produces stable motion vectors.

Less noise.

More reliable for large/general motion.

Misses small movements.

Method B

Detects fine and small movements.

More sensitive to motion details.

Produces noisy vectors.

Less stable.

For dynamic scenes, Method A is generally more suitable when stability and reliable tracking are the main requirements.

Method B is better when detecting small or fine movements is more important and some noise can be tolerated.

Conclusion

Accuracy of fine motion: Method B

Stability: Method A

Noise sensitivity: Method B is more sensitive

General dynamic-scene tracking: Method A

4. Model Generalization and Overfitting

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Model A

Training accuracy = 98%

Testing accuracy = 70%

Large difference:

98−70=28%

This indicates overfitting. Model A performs extremely well on training data but poorly on unseen data.

Model B

Training accuracy = 90%

Testing accuracy = 88%

Difference:

90−88=2%

This indicates good generalization and low overfitting.

Model B should be selected for real-world deployment because its training and testing accuracies are close.

Conclusion: Model B is more reliable because it generalizes better to unseen data, while Model A has a high risk of overfitting.

5. System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate

C	95%	120 ms	High

Model A

Lowest latency: 50 ms

Lowest cost

Lowest accuracy: 88%

Good for highly time-sensitive applications.

Model B

Accuracy: 92%

Latency: 80 ms

Moderate cost

Good balance between performance and resources.

Model C

Highest accuracy: 95%

Highest latency: 120 ms

Highest cost

May not be suitable for strict real-time applications.

Model B is the best overall choice because it provides:

Good accuracy (92%)

Reasonable latency (80 ms)

Moderate cost

Model A is preferable if very low latency and low cost are the highest priorities. Model C is preferable only when maximum accuracy is more important than latency and cost.

Final Conclusion

For a practical real-time application, Model B provides the best balance between accuracy, latency, and cost.